ACCEPTED | Applicant 209765

SHAP and LIME explain the same LightGBM input and the same default-class output

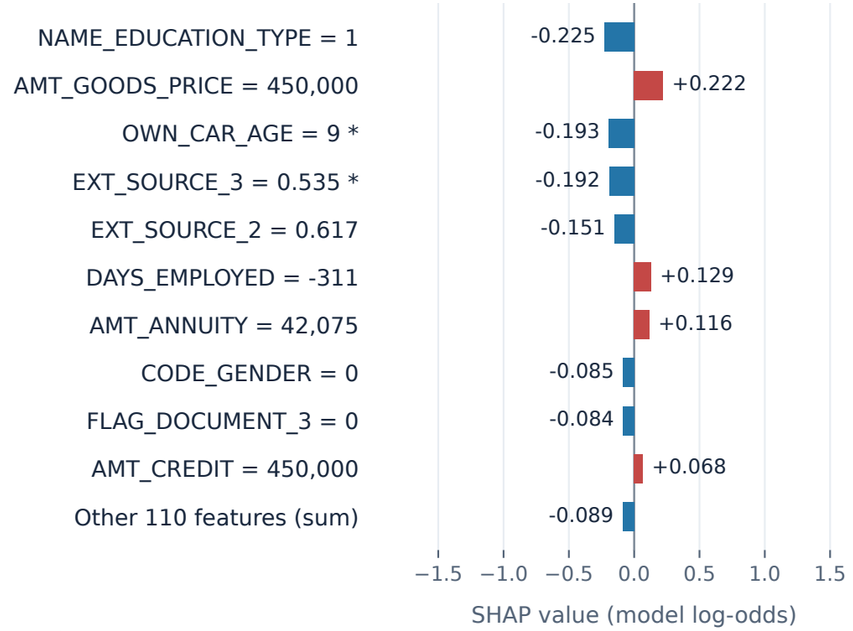
Model score 0.274

Accept < 0.500 | Reject >= 0.500

SHAP

Feature value + exact TreeSHAP attribution

Top-10 attribution coverage: 69.3% | Raw-input score: 0.454

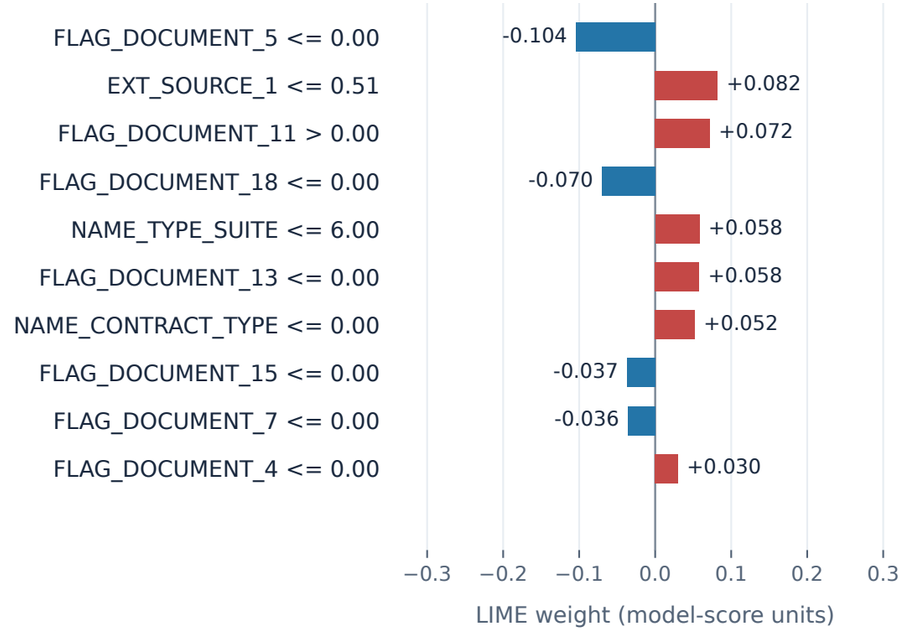


Blue / negative: lowers default-class score

LIME

Discretized condition + local surrogate weight

Local fidelity R^2: 0.061 | 1,000 perturbations | seed 42



Red / positive: raises default-class score

EXACT MODEL RECONSTRUCTION

Base -0.493 + all contributions -0.483 = -0.976

Logistic transform -> 0.274 model score

LOCAL APPROXIMATION CHECK

Surrogate 0.397 | Model 0.274 | Gap 0.123

Condition boundaries describe bins; they are not approval thresholds.

Same median-filled input for both methods; * = imputed value. Top 10 ranked within each method. Panel units differ.

Recomputed from paper code. Acceptance/rejection is a model rule, not a recorded lending decision. Score is uncalibrated.

ACCEPTED | Applicant 342011

SHAP and LIME explain the same LightGBM input and the same default-class output

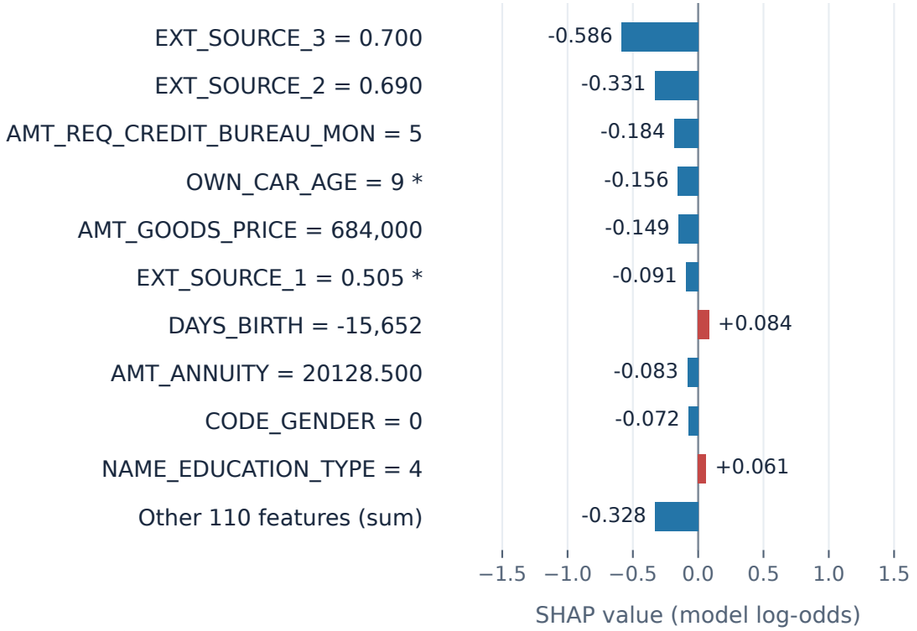
Model score 0.089

Accept < 0.500 | Reject >= 0.500

SHAP

Feature value + exact TreeSHAP attribution

Top-10 attribution coverage: 68.1% | Raw-input score: 0.141

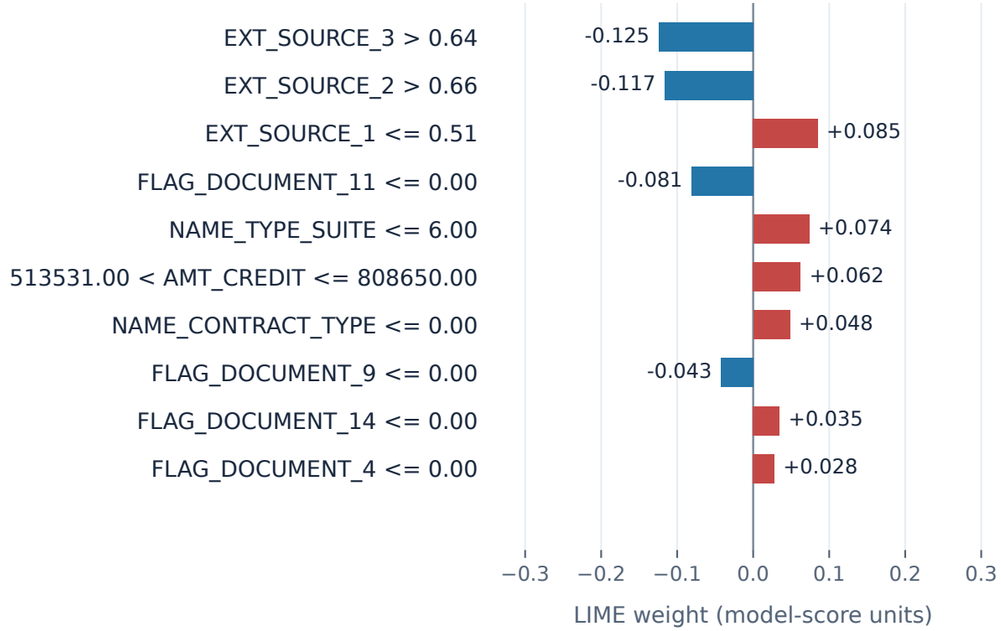


Blue / negative: lowers default-class score

LIME

Discretized condition + local surrogate weight

Local fidelity R^2: 0.293 | 1,000 perturbations | seed 42



Red / positive: raises default-class score

EXACT MODEL RECONSTRUCTION

Base -0.493 + all contributions -1.835 = -2.328

Logistic transform -> 0.089 model score

LOCAL APPROXIMATION CHECK

Surrogate 0.199 | Model 0.089 | Gap 0.110

Condition boundaries describe bins; they are not approval thresholds.

Same median-filled input for both methods; * = imputed value. Top 10 ranked within each method. Panel units differ.

Recomputed from paper code. Acceptance/rejection is a model rule, not a recorded lending decision. Score is uncalibrated.

REJECTED | Applicant 218332

SHAP and LIME explain the same LightGBM input and the same default-class output

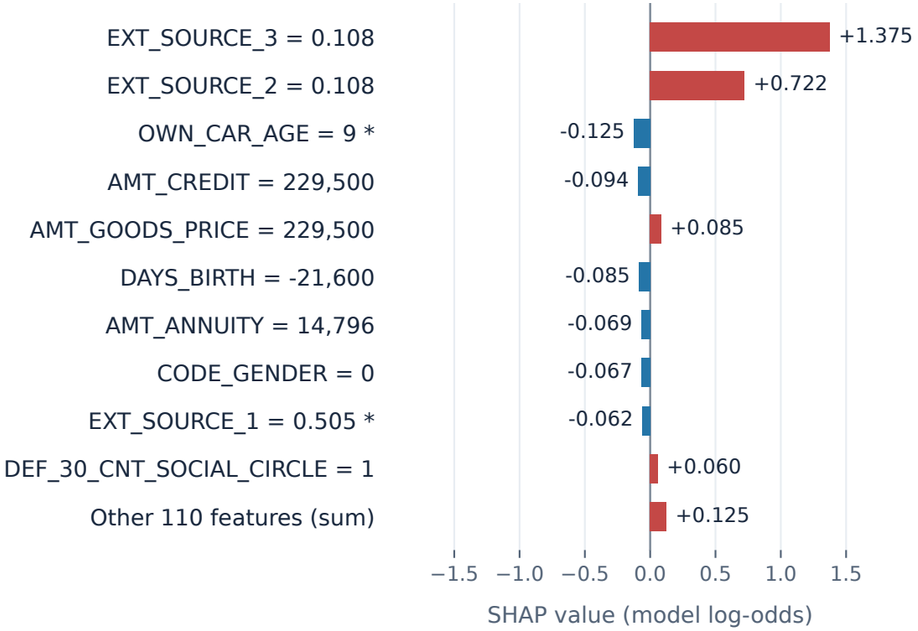
Model score 0.798

Accept < 0.500 | Reject >= 0.500

SHAP

Feature value + exact TreeSHAP attribution

Top-10 attribution coverage: 83.4% | Raw-input score: 0.814

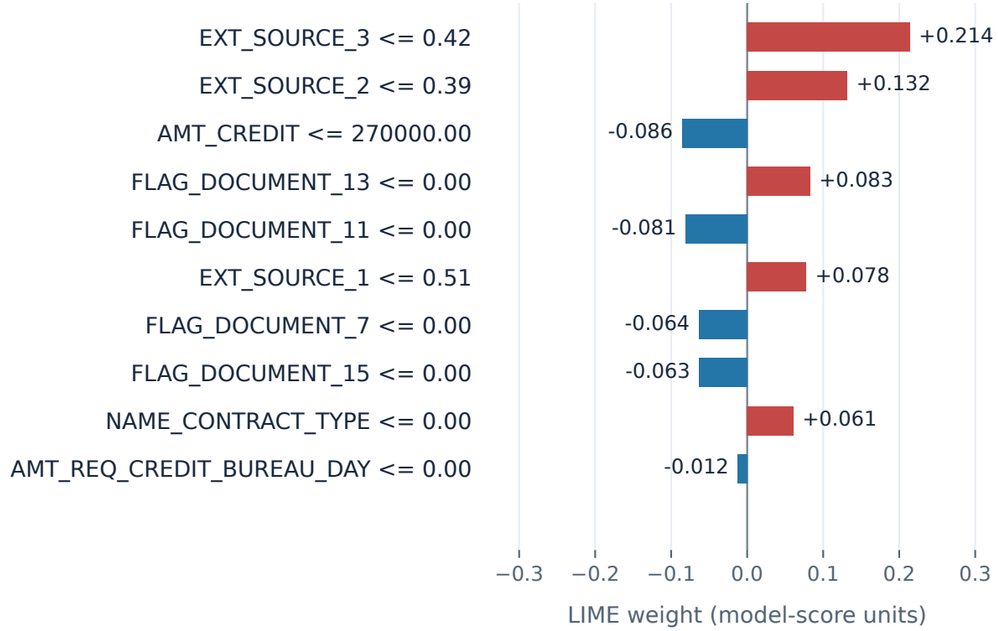


Blue / negative: lowers default-class score

LIME

Discretized condition + local surrogate weight

Local fidelity R^2: 0.549 | 1,000 perturbations | seed 42



Red / positive: raises default-class score

EXACT MODEL RECONSTRUCTION

Base -0.493 + all contributions +1.865 = +1.373

Logistic transform -> 0.798 model score

LOCAL APPROXIMATION CHECK

Surrogate 0.526 | Model 0.798 | Gap 0.272

Condition boundaries describe bins; they are not approval thresholds.

Same median-filled input for both methods; * = imputed value. Top 10 ranked within each method. Panel units differ.

Recomputed from paper code. Acceptance/rejection is a model rule, not a recorded lending decision. Score is uncalibrated.

REJECTED | Applicant 425755

SHAP and LIME explain the same LightGBM input and the same default-class output

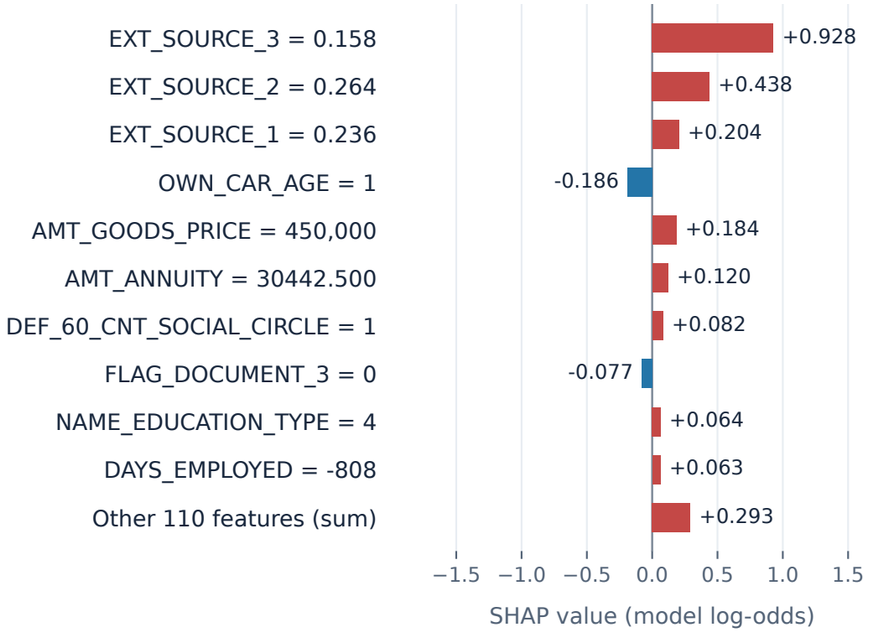
Model score 0.835

Accept < 0.500 | Reject >= 0.500

SHAP

Feature value + exact TreeSHAP attribution

Top-10 attribution coverage: 80.3% | Raw-input score: 0.821

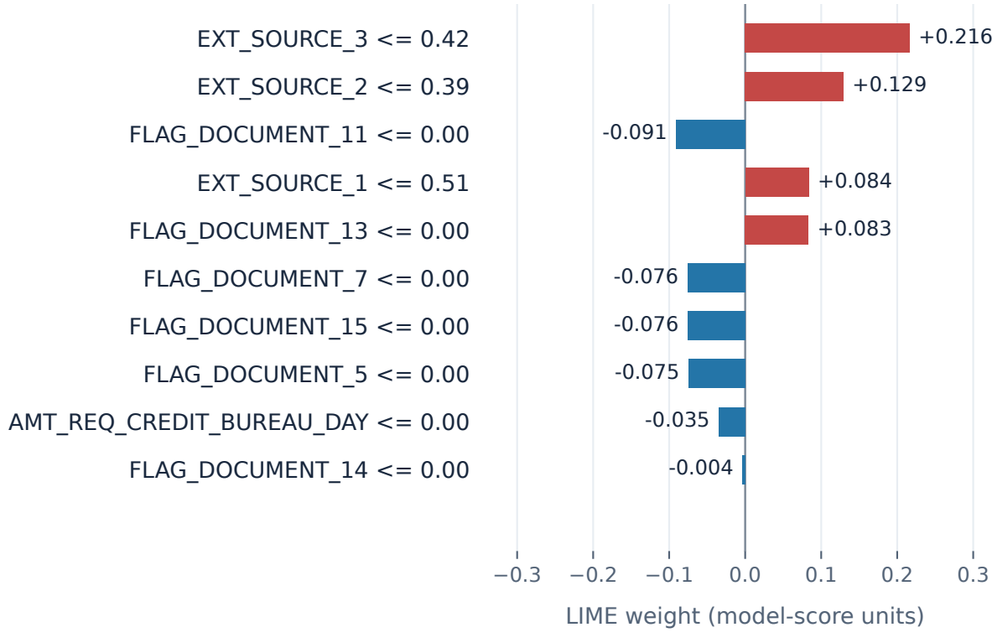


Blue / negative: lowers default-class score

LIME

Discretized condition + local surrogate weight

Local fidelity R^2: 0.489 | 1,000 perturbations | seed 42



Red / positive: raises default-class score

EXACT MODEL RECONSTRUCTION

Base -0.493 + all contributions +2.113 = +1.621

Logistic transform -> 0.835 model score

LOCAL APPROXIMATION CHECK

Surrogate 0.583 | Model 0.835 | Gap 0.252

Condition boundaries describe bins; they are not approval thresholds.

Same median-filled input for both methods; * = imputed value. Top 10 ranked within each method. Panel units differ.

Recomputed from paper code. Acceptance/rejection is a model rule, not a recorded lending decision. Score is uncalibrated.